

Online Learning

Online Gradient Descent & Subgradients

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Credits for the resource

- The slides are based on
 - the lectures of Prof. Luca Trevisan:
<https://lucatrevisan.github.io/40391/index.html>
 - the lectures of Prof. Shipra Agrawal:
<https://ieor8100.github.io/mab/>
 - the lectures of Prof. Francesco Orabona:
<https://parameterfree.com/lecture-notes-on-online-learning/>
 - the monograph: <https://arxiv.org/abs/1912.13213>
 - Elad Hazan's textbook:
Introduction to Online Convex Optimization, 2nd Edition.



Outline

- 1 Does FTL always work?
- 2 Gradient Descent for Online Convex Optimization (GD)
- 3 Subgradient & Subdifferential



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- First, we assume to make no assumptions on $\mathcal{K}$ and $\{f_t : L \mapsto \mathbb{R}\}$.
- At time t , we are given previous cost functions $f_1, \dots, f_{t-1}$, and then give the solution

$$\mathbf{x}_t := \arg \min_{\mathbf{x} \in \mathcal{K}} \sum_{k=1}^{t-1} f_k(\mathbf{x}).$$



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That is, the best solution for the previous $t - 1$ steps.



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That is, the best solution for the previous $t-1$ steps.

- It seems reasonable and makes sense, doesn't it?



FTL leads to “overfitting”

$$t: 1$$

$$\mathbf{x}_t: (0.5, 0.5)$$

$$\ell_t: (0, 0.5)$$

$$f_t(\mathbf{x}_t): 0.25$$

$$\arg \min_{\mathbf{x}} \sum_{k=1}^t f_k(\mathbf{x}): (1, 0)$$

FTL leads to “overfitting”

t :	1	2
$\mathbf{x}_t$:	(0.5, 0.5)	(1, 0)
ℓ_t :	(0, 0.5)	(1, 0)
$f_t(\mathbf{x}_t)$:	0.25	1
$\arg \min_{\mathbf{x}} \sum_{k=1}^t f_k(\mathbf{x})$:	(1, 0)	(0, 1)

FTL leads to “overfitting”

t :	1	2	3
$\mathbf{x}_t$:	(0.5, 0.5)	(1, 0)	(0, 1)
ℓ_t :	(0, 0.5)	(1, 0)	(0, 1)
$f_t(\mathbf{x}_t)$:	0.25	1	1
$\arg \min_{\mathbf{x}} \sum_{k=1}^t f_k(\mathbf{x})$:	(1, 0)	(0, 1)	(1, 0)

FTL leads to “overfitting”

t :	1	2	3	4
$\mathbf{x}_t$:	(0.5, 0.5)	(1, 0)	(0, 1)	(1, 0)
ℓ_t :	(0, 0.5)	(1, 0)	(0, 1)	(1, 0)
$f_t(\mathbf{x}_t)$:	0.25	1	1	1
$\arg \min_{\mathbf{x}} \sum_{k=1}^t f_k(\mathbf{x})$:	(1, 0)	(0, 1)	(1, 0)	(0, 1)

FTL leads to “overfitting”

t :	1	2	3	4	5
$\mathbf{x}_t$:	(0.5, 0.5)	(1, 0)	(0, 1)	(1, 0)	(0, 1)
ℓ_t :	(0, 0.5)	(1, 0)	(0, 1)	(1, 0)	(0, 1)
$f_t(\mathbf{x}_t)$:	0.25	1	1	1	1
$\arg \min_{\mathbf{x}} \sum_{k=1}^t f_k(\mathbf{x})$:	(1, 0)	(0, 1)	(1, 0)	(0, 1)	(1, 0)

FTL leads to “overfitting”

t :	1	2	3	4	5	...
$\mathbf{x}_t$:	(0.5, 0.5)	(1, 0)	(0, 1)	(1, 0)	(0, 1)	...
ℓ_t :	(0, 0.5)	(1, 0)	(0, 1)	(1, 0)	(0, 1)	...
$f_t(\mathbf{x}_t)$:	0.25	1	1	1	1	...
$\arg \min_{\mathbf{x}} \sum_{k=1}^t f_k(\mathbf{x})$:	(1, 0)	(0, 1)	(1, 0)	(0, 1)	(1, 0)	...

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t :	1	2	3	4	5	...
$\mathbf{x}_t$:	(0.5, 0.5)	(1, 0)	(0, 1)	(1, 0)	(0, 1)	...
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$f_t(\mathbf{x}_t)$:	0.25	1	1	1	1	...
$\arg \min_{\mathbf{x}} \sum_{k=1}^t f_k(\mathbf{x})$:	(1, 0)	(0, 1)	(1, 0)	(0, 1)	(1, 0)	...

optimum loss: $\approx T/2$.

FTL's loss: $\approx T$.

regret: $\approx T/2$ (linear).



Remark

- Note that the first example of no-regret analysis in this course uses a special kind of loss function.
 - Squared difference: $\|\mathbf{x}_t - \mathbf{y}_t\|_2^2$.



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Online Gradient Descent (GD)

- ① **Input:** convex set $\mathcal{K}$, T , $\mathbf{x}_1 \in \mathcal{K}$, learning rate $\{\eta_t\}$.
- ② **for** $t \leftarrow 1$ to T **do**:
 - ① Play $\mathbf{x}_t$ and observe cost $f_t(\mathbf{x}_t)$.
 - ② Update and Project:

$$\mathbf{y}_{t+1} = \mathbf{x}_t - \eta_t \nabla f_t(\mathbf{x}_t)$$

$$\mathbf{x}_{t+1} = \Pi_{\mathcal{K}}(\mathbf{y}_{t+1})$$

- ③ **end for**



GD for online convex optimization is of no-regret

Theorem A

Online gradient descent with learning rate $\{\eta_t = \frac{D}{G\sqrt{t}}, t \in [T]\}$ guarantees the following for all $T \geq 1$:

$$\text{regret}_T = \sum_{t=1}^T f_t(\mathbf{x}_t) - \min_{\mathbf{x}^* \in \mathcal{K}} \sum_{t=1}^T f_t(\mathbf{x}^*) \leq \frac{3}{2}GD\sqrt{T}.$$

- D : the diameter of $\mathcal{K}$.
- Assume that $\nabla f_t(\mathbf{x}) \leq G$ for each $\mathbf{x} \in \mathcal{K}$.



Proof of Theorem A (1/4)

- Let $\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{K}} \sum_{t=1}^T f_t(\mathbf{x})$.
- Since f_t is convex, we have

$$f_t(\mathbf{x}_t) - f_t(\mathbf{x}^*) \leq (\nabla f_t(\mathbf{x}_t))^\top (\mathbf{x}_t - \mathbf{x}^*).$$

- By the updating rule for $\mathbf{x}_{t+1}$ and the Pythagorean theorem, we have

$$\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 = \|\Pi_{\mathcal{K}}(\mathbf{x}_t - \eta_t \nabla f_t(\mathbf{x}_t)) - \mathbf{x}^*\|^2 \leq \|\mathbf{x}_t - \eta_t \nabla f_t(\mathbf{x}_t) - \mathbf{x}^*\|^2.$$



Proof of Theorem A (2/4)

- Hence

$$\begin{aligned}\|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2 &\leq \|\mathbf{x}_t - \mathbf{x}^*\|^2 + \eta_t^2 \|\nabla f_t(\mathbf{x}_t)\|^2 - 2\eta_t (\nabla f_t(\mathbf{x}_t))^\top (\mathbf{x}_t - \mathbf{x}^*) \\ 2(\nabla f_t(\mathbf{x}_t))^\top (\mathbf{x}_t - \mathbf{x}^*) &\leq \frac{\|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2}{\eta_t} + \eta_t G^2.\end{aligned}$$

- Summing above inequality from $t = 1$ to T and setting $\eta_t = \frac{D}{G\sqrt{t}}$ and $\frac{1}{\eta_0} := 0$ we have :



Proof of Theorem A (3/4)

$$\begin{aligned} 2 \left(\sum_{t=1}^T f_t(\mathbf{x}_t) - f_t(\mathbf{x}^*) \right) &\leq 2 \sum_{t=1}^T \nabla f_t(\mathbf{x}_t)^\top (\mathbf{x}_t - \mathbf{x}^*) \\ &\leq \sum_{t=1}^T \frac{\|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2}{\eta_t} + G^2 \sum_{t=1}^T \eta_t \\ &\leq \sum_{t=1}^T \|\mathbf{x}_t - \mathbf{x}^*\|^2 \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} \right) + G^2 \sum_{t=1}^T \eta_t \\ &\leq D^2 \sum_{t=1}^T \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} \right) + G^2 \sum_{t=1}^T \eta_t \\ &\leq D^2 \frac{1}{\eta_T} + G^2 \sum_{t=1}^T \eta_t \\ &\leq 3DG\sqrt{T}. \end{aligned}$$



Proof of Theorem A (4/4)

Note that we can also deduce in this way

$$\sum_{t=1}^T \frac{\|\mathbf{x}_t - \mathbf{x}^*\|^2 - \|\mathbf{x}_{t+1} - \mathbf{x}^*\|^2}{\eta_t} + G^2 \sum_{t=1}^T \eta_t$$



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 = & \frac{\|\mathbf{x}_1 - \mathbf{x}^*\|^2}{\eta_1} + \|\mathbf{x}_2 - \mathbf{x}^*\| \left(\frac{1}{\eta_2} - \frac{1}{\eta_1} \right) + \|\mathbf{x}_3 - \mathbf{x}^*\| \left(\frac{1}{\eta_3} - \frac{1}{\eta_2} \right) + \dots \\
 & + \|\mathbf{x}_T - \mathbf{x}^*\| \left(\frac{1}{\eta_T} - \frac{1}{\eta_{T-1}} \right) - \frac{\|\mathbf{x}_T - \mathbf{x}^*\|}{\eta_T} + G^2 \sum_{t=1}^T \eta_t \\
 \leq & \frac{\|\mathbf{x}_1 - \mathbf{x}^*\|^2}{\eta_1} + D^2 \sum_{t=2}^T \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} \right) + G^2 \sum_{t=1}^T \eta_t
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 \leq & \frac{\|\mathbf{x}_1 - \mathbf{x}^*\|^2}{\eta_1} + D^2 \sum_{t=2}^T \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} \right) + G^2 \sum_{t=1}^T \eta_t \\
 \leq & \frac{D^2}{\eta_1} + D^2 \left(\frac{1}{\eta_T} - \frac{1}{\eta_1} \right) + G^2 \sum_{t=1}^T \eta_t \\
 = & \frac{D^2}{\eta_T} + G^2 \sum_{t=1}^T \frac{D}{G\sqrt{t}} \leq
 \end{aligned}$$



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 \leq & \frac{\|\mathbf{x}_1 - \mathbf{x}^*\|^2}{\eta_1} + D^2 \sum_{t=2}^T \left(\frac{1}{\eta_t} - \frac{1}{\eta_{t-1}} \right) + G^2 \sum_{t=1}^T \eta_t \\
 \leq & \frac{D^2}{\eta_1} + D^2 \left(\frac{1}{\eta_T} - \frac{1}{\eta_1} \right) + G^2 \sum_{t=1}^T \eta_t \\
 = & \frac{D^2}{\eta_T} + G^2 \sum_{t=1}^T \frac{D}{G\sqrt{t}} \leq \frac{D^2}{D/(G\sqrt{T})} + DG(2\sqrt{T} - 1)
 \end{aligned}$$



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 \leq & \frac{D^2}{\eta_1} + D^2 \left(\frac{1}{\eta_T} - \frac{1}{\eta_1} \right) + G^2 \sum_{t=1}^T \eta_t \\
 = & \frac{D^2}{\eta_T} + G^2 \sum_{t=1}^T \frac{D}{G\sqrt{t}} \leq \frac{D^2}{D/(G\sqrt{T})} + DG(2\sqrt{T} - 1) \leq 3DG\sqrt{T}.
 \end{aligned}$$



The Lower Bound (for OLO)

Theorem B

Let $\mathcal{K} = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_\infty \leq r\}$ be a convex subset of $\mathbb{R}^d$. Let A be any algorithm for Online **Linear** Optimization on $\mathcal{K}$. Then for any $T \geq 1$, there exists a sequence of vectors $\mathbf{g}_1, \dots, \mathbf{g}_T$ with $\|\mathbf{g}_t\|_2 \leq L$ and $\mathbf{u} \in \mathcal{K}$ such that the regret of A satisfies

$$\text{regret}_T(\mathbf{u}) = \sum_{t=1}^T \langle \mathbf{g}_t, \mathbf{x}_t \rangle - \sum_{t=1}^T \langle \mathbf{g}_t, \mathbf{u} \rangle \geq \frac{\sqrt{2}LD\sqrt{T}}{4}.$$

- The diameter D of $\mathcal{K}$ is at most $\sqrt{\sum_{i=1}^d (2r)^2} \leq 2r\sqrt{d}$.
- $\|\mathbf{x}\|_\infty \leq r \Leftrightarrow |\mathbf{x}(i)| \leq r$ for each $i \in [n]$.



Proof of Theorem B (1/2)

- The approach:

For any random variable $\mathbf{z}$ with domain $\mathcal{V}$ and any function f ,

$$\sup_{\mathbf{x} \in V} f(\mathbf{x}) \geq E[f(\mathbf{z})].$$



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- $\text{regret}_T = \max_{\mathbf{u} \in \mathcal{K}} \text{regret}_T(\mathbf{u})$.



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- $\text{regret}_T = \max_{\mathbf{u} \in \mathcal{K}} \text{regret}_T(\mathbf{u})$.
- Let $\mathbf{v}, \mathbf{w} \in \mathcal{K}$ such that $\|\mathbf{v} - \mathbf{w}\| = D$.



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- Let $\mathbf{v}, \mathbf{w} \in \mathcal{K}$ such that $\|\mathbf{v} - \mathbf{w}\| = D$.
- Let $\mathbf{z} := \frac{\mathbf{v} - \mathbf{w}}{\|\mathbf{v} - \mathbf{w}\|}$



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- Let $\mathbf{z} := \frac{\mathbf{v} - \mathbf{w}}{\|\mathbf{v} - \mathbf{w}\|} \Rightarrow \langle \mathbf{z}, \mathbf{v} - \mathbf{w} \rangle = D$.
- Let $\epsilon_1, \epsilon_2, \dots, \epsilon_T$ be i.i.d. random variables such that $\Pr[\epsilon_t = 1] = \Pr[\epsilon_t = -1] = 1/2$ for each t .



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- We choose the losses $\mathbf{g}_t = L\epsilon_t \mathbf{z}$.



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- Let $\epsilon_1, \epsilon_2, \dots, \epsilon_T$ be i.i.d. random variables such that $\Pr[\epsilon_t = 1] = \Pr[\epsilon_t = -1] = 1/2$ for each t .
- We choose the losses $\mathbf{g}_t = L\epsilon_t \mathbf{z}$.
 - The cost at t : $\langle L\epsilon_t \mathbf{z}, \mathbf{x}_t \rangle$.
 - $\|\mathbf{g}_t\| = \sqrt{L^2 \epsilon_t^2} \cdot \|\mathbf{z}\| \leq L$.



Proof of Theorem B (2/2)

$$\begin{aligned}
 \sup_{\mathbf{g}_1, \dots, \mathbf{g}_T} \text{regret}_T &\geq E \left[\sum_{t=1}^T L\epsilon_t \langle \mathbf{z}, \mathbf{x}_t \rangle - \min_{\mathbf{u} \in \mathcal{K}} \sum_{t=1}^T L\epsilon_t \langle \mathbf{z}, \mathbf{u} \rangle \right] \\
 &= E \left[- \min_{\mathbf{u} \in \mathcal{K}} \sum_{t=1}^T L\epsilon_t \langle \mathbf{z}, \mathbf{u} \rangle \right] = E \left[\max_{\mathbf{u} \in \mathcal{K}} \sum_{t=1}^T L\epsilon_t \langle \mathbf{z}, \mathbf{u} \rangle \right] \\
 &\geq E \left[\max_{\mathbf{u} \in \{\mathbf{v}, \mathbf{w}\}} \sum_{t=1}^T L\epsilon_t \langle \mathbf{z}, \mathbf{u} \rangle \right] \\
 &= E \left[\frac{1}{2} \sum_{t=1}^T L\epsilon_t \langle \mathbf{z}, \mathbf{v} + \mathbf{w} \rangle + \frac{1}{2} \left| \sum_{t=1}^T L\epsilon_t \langle \mathbf{z}, \mathbf{v} - \mathbf{w} \rangle \right| \right] \\
 &\geq \frac{L}{2} E \left[\left| \sum_{t=1}^T \epsilon_t \langle \mathbf{z}, \mathbf{v} - \mathbf{w} \rangle \right| \right] = \frac{LD}{2} E \left[\left| \sum_{t=1}^T \epsilon_t \right| \right] \\
 &\geq \frac{\sqrt{2}LD\sqrt{T}}{4}. \quad (\text{by Khintchine inequality})
 \end{aligned}$$



Exercise

Prove the last inequality by Khintchine inequality.



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 - $f_t(\mathbf{x}) = \max(1 - \langle \mathbf{z}, \mathbf{x} \rangle, 0)$ for $\mathbf{x}, \mathbf{z} \in \mathbb{R}^d$.
 - ReLU activation function: $f_t(x) = \max(x, 0)$.



Recall

Subgradient

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$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \mathbf{g}, \mathbf{y} - \mathbf{x} \rangle.$$



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- If f is convex, then $\partial f(\mathbf{x})$ turns out to be $\nabla f(\mathbf{x})$.



Examples and Useful Facts (1/5)

Theorem

Let $f_1, f_2, \dots, f_m$ be convex functions on $\mathbb{R}^d$ and let $f = f_1 + f_2 + \dots + f_m$. Then $\partial f(\mathbf{x}) \supseteq \partial f_1(\mathbf{x}) + \partial f_2(\mathbf{x}) + \dots + \partial f_m(\mathbf{x})$, for each $\mathbf{x} \in \mathbb{R}^d$.

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Examples and Useful Facts (2/5)

Subgradients of an Absolute Value Function

Let $f(x) = |x|$, then the subdifferential set $\partial f(x)$ is

$$\partial f(x) = \begin{cases} \{1\}, & x > 0, \\ \{-1\}, & x < 0, \\ [-1, 1], & x = 0. \end{cases}$$



Examples and Useful Facts (3/5)

Subgradients of the Hinge Loss

Consider $f : \mathbb{R}^d \mapsto \mathbb{R}$, such that $f(\mathbf{x}) = \max(1 - \langle \mathbf{z}, \mathbf{x} \rangle, 0)$ for $\mathbf{z} \in \mathbb{R}^d$. Then the subdifferential set $\partial f(\mathbf{x})$ is

$$\partial f(\mathbf{x}) = \begin{cases} \{\mathbf{0}\}, & \text{if } 1 - \langle \mathbf{z}, \mathbf{x} \rangle < 0 \\ \{-\alpha \mathbf{z} : \alpha \in [0, 1]\}, & \text{if } 1 - \langle \mathbf{z}, \mathbf{x} \rangle = 0 \\ \{-\mathbf{z}\}, & \text{otherwise} \end{cases}.$$



Examples and Useful Facts (4/5)

Subgradients of 2-Norm

Consider $f : \mathbb{R}^d \mapsto \mathbb{R}$, such that $f(\mathbf{x}) = \|\mathbf{x}\|_2$. Then the subdifferential set $\partial f(\mathbf{x})$ is

$$\partial f(\mathbf{x}) = \begin{cases} \mathbf{x}/\|\mathbf{x}\|_2, & \text{for } \mathbf{x} \neq \mathbf{0}, \\ \{\mathbf{z} : \|\mathbf{z}\|_2 \leq 1\}, & \text{for } \mathbf{x} = \mathbf{0}. \end{cases}$$



Examples and Useful Facts (5/5)

Lipschitz Continuity

A function $f : \mathbb{R}^d \mapsto \mathbb{R}$ is **L -Lipschitz** over a set V with respect to a norm $\|\cdot\|$ if $|f(\mathbf{x}) - f(\mathbf{y})| \leq L\|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in V$.

Theorem

Let $f : \mathbb{R}^d \mapsto \mathbb{R}$ be a convex function. Then, f is L -Lipschitz in $\text{int dom}(f)$ with respect to the L_2 -norm if and only if

for all $\mathbf{x} \in \text{int dom}(f)$ and $\mathbf{g} \in \partial f(\mathbf{x})$, we have $\|\mathbf{g}\|_2 \leq L$.

★ The (sub-)gradient is also bounded by L !



A Recall for Conventional Continuity

Continuous Function

For a function $f : D \mapsto \mathbb{R}$, $D \subseteq \mathbb{R}$, f is continuous at $x_0 \in D$ if and only if

$$\forall \epsilon > 0, \exists \delta > 0, \text{ s.t. for all } x \in D, |x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \epsilon.$$



Proof

- Assume that f is L -Lipschitz, then



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- Assume that f is L -Lipschitz, then $|f(\mathbf{x}) - f(\mathbf{y})| \leq L\|\mathbf{x} - \mathbf{y}\|_2$, for each $\mathbf{x}, \mathbf{y} \in \text{dom}(f)$.



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- Assume that f is L -Lipschitz, then $|f(\mathbf{x}) - f(\mathbf{y})| \leq L\|\mathbf{x} - \mathbf{y}\|_2$, for each $\mathbf{x}, \mathbf{y} \in \text{dom}(f)$.
- For small enough $\epsilon > 0$, let $\mathbf{y} = \mathbf{x} + \epsilon \frac{\mathbf{g}}{\|\mathbf{g}\|_2} \in \text{int dom}(f)$.
 - $\|\mathbf{x} - \mathbf{y}\|_2 =$



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- Then,

$$L\epsilon = L\|\mathbf{x} - \mathbf{y}\|_2 \geq |f(\mathbf{x}) - f(\mathbf{y})|$$



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Projected Online Subgradient Descent

- ① **Input:** convex set $\mathcal{K}$, T , $\mathbf{x}_1 \in \mathcal{K}$, step size $\{\eta_t\}$.
- ② **for** $t \leftarrow 1$ to T **do**:
 - ① Play $\mathbf{x}_t$ and observe cost $f_t(\mathbf{x}_t)$.
 - ② Set $\mathbf{g}_t \in \partial f_t(\mathbf{x}_t)$.
 - ③ Update and Project:

$$\mathbf{y}_{t+1} = \mathbf{x}_t - \eta_t \mathbf{g}_t$$

$$\mathbf{x}_{t+1} = \Pi_{\mathcal{K}}(\mathbf{y}_{t+1})$$

- ③ **end for**



Discussions

